

جامعة الخرطوم



كلية الهندسة – قسم الهندسة الكهربائية والإلكترونية

امتحانات الصف الثالث لنيل درجة بكالوريوس الشرف في الهندسة الكهربائية والإلكترونية

الفصل الدراسي الثاني للعام الدراسي 2022 - 2023

اسم المقرر: البرمجة بلغة جافا **Java programming** - رقم المقرر: **EEE 32502**

التاريخ: نوفمبر 2023	عدد الصفحات 3	الزمن: ساعتان
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Question 1: briefly answer the following questions.

- 1.1 What do we mean by saying that Java is architecture-neutral language?
- 1.2 How did the World Wide Web and the internet participate in the programming revolution?
- 1.3 What is the main difference between OOP and POP?
- 1.4 What are the components of the JRE?
- 1.5 What is the difference between inheritance and polymorphism?
- 1.6 What do we mean by encapsulation?
- 1.7 There are two ways to achieve abstraction. Mention them.
- 1.8 A thread could be terminated due to three reasons. Mention them.
- 1.9 What are the three default streams in Java?
- 1.10 On which API Java swing was built on top of?

Question 2: state if the sentence is true or false and correct the false statements.

2.1 The C++ language was called “C with classes”.

2.2 It is not possible for an interface to inherit another interface's properties.

2.3 The Java byte code is a .class file that is executed by the Java run-time system.

2.4 Run time polymorphism could be achieved by method overloading.

2.5 Abstract classes could have non-abstract methods within them.

2.6 Java threads are lightweight threads.

2.7 There is one JDBC driver for all database sources.

Question 3: fill in the blanks.

3.1 _____ is one of the reasons that computer language innovation occurred.

3.2 _____ is an example of a POP language.

3.3 POP does not allow _____ flow freely in the program.

3.4 _____ are small programs that need a java enabled web browser to be executed.

3.5 a _____ is a component which can contain other elements like buttons.

3.6 The model- view- _____ architecture is used in Swing.

3.7 _____ is a Java API to connect and interact with databases.

Question 4: For each block code, complete the **CODE** to get the output and provide explanation for your answer:

4.1

```
public class Exams {  
    public static void main(String[] args) {  
  
        String[] seats = {"M", "C", "G", "P", "D"};  
        System.out.println("It's time to take your 3th years exams. Please, take your seats.\n");  
        for (int i = 0; i < seats.length; i++) {  
            System.out.println(CODE);  
        }  
    }  
}
```

Output:

It's time to take your 3th years exams. Please, take your seats.

M, you will take seat 0
C, you will take seat 1
G, you will take seat 2
P, you will take seat 3
D, you will take seat 4

4.2

```
public class Agree {  
    public static void main(String[] args) {  
        int totalVotes = 760;  
        System.out.println("Statement: If Java was easy, they would call it Python!");  
        CODE  
    }  
    public static void calculateVotes(double numOfVotes) {  
        // 15% of votes do agree with the statement.  
  
        double portion = numOfVotes*0.15;  
  
        System.out.println("The number of people who agree with this statement is " + (int) portion);  
    }  
}
```

Output:

Statement: If Java was easy, they would call it Python!
The number of people who agree with this statement is 114